

Computer Vision and Multiview Geometry

DT084A Lab - 4 Report

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Task 1: K-means Clustering

K-means clustering minimizes the within-cluster sum of squared distances between the data points and the corresponding cluster centroids. The objective function is defined as

$$J = \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

where

- K is the number of clusters
- C_k is the set of points assigned to cluster k
- x_i is a data point
- μ_k is the centroid of cluster k

The initial cluster centers are given as

$$\mu_1^{(0)} = (1, 1.5)$$

$$\mu_2^{(0)} = (3, 1)$$

The dataset used in this example is

$$X = \{(1, 1), (1, 2), (2, 1), (3, 1), (3, 2), (4, 1)\}$$

Distances are computed using the squared Euclidean distance

$$d(x, \mu) = (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2$$

Iteration 1

Assignment Step

Each point is assigned to the closest cluster center.

Point	Distance to C_1	Distance to C_2	Cluster
(1,1)	0.25	4	C_1
(1,2)	0.25	5	C_1
(2,1)	1.25	1	C_2
(3,1)	4.25	0	C_2
(3,2)	4.25	1	C_2
(4,1)	9.25	1	C_2

The resulting clusters are

$$C_1 = \{(1, 1), (1, 2)\}$$

$$C_2 = \{(2, 1), (3, 1), (3, 2), (4, 1)\}$$

Update Step

The new cluster centers are computed as the mean of the points assigned to each cluster.

Cluster 1:

$$\mu_1^{(1)} = \left(\frac{1+1}{2}, \frac{1+2}{2} \right) = (1, 1.5)$$

Cluster 2:

$$\mu_2^{(1)} = \left(\frac{2+3+3+4}{4}, \frac{1+1+2+1}{4} \right) = (3, 1.25)$$

Iteration 2

Using the updated centers

$$\mu_1^{(1)} = (1, 1.5)$$

$$\mu_2^{(1)} = (3, 1.25)$$

Distances are recomputed. The assignments remain unchanged:

$$C_1 = \{(1, 1), (1, 2)\}$$

$$C_2 = \{(2, 1), (3, 1), (3, 2), (4, 1)\}$$

Update Step

Cluster 1:

$$\mu_1^{(2)} = (1, 1.5)$$

Cluster 2:

$$\mu_2^{(2)} = (3, 1.25)$$

Since the centers do not change anymore, the algorithm has converged. Final clusters:

$$C_1 = \{(1, 1), (1, 2)\}$$

$$C_2 = \{(2, 1), (3, 1), (3, 2), (4, 1)\}$$

Final centroids:

$$\mu_1 = (1, 1.5)$$

$$\mu_2 = (3, 1.25)$$

Task 2: Traditional Classification Algorithms in Computer Vision

Traditional computer vision pipelines typically consist of two stages: feature extraction (e.g., HOG, SIFT, LBP) followed by a machine learning classifier.

Algorithm	When Applicable	Advantages	Disadvantages
k-Nearest Neighbors (k-NN)	Small or medium datasets, simple classification problems	Very simple implementation, no training phase, flexible decision boundaries	Slow prediction time, sensitive to noise and feature scaling
Support Vector Machine (SVM)	High-dimensional feature spaces such as HOG or SIFT descriptors	Strong performance, robust to overfitting, effective in high dimensions	Parameter tuning required, computationally expensive for large datasets
Decision Trees	Datasets where interpretability is important	Easy to interpret, handles nonlinear relationships	Prone to overfitting, unstable to small data changes
Random Forest	Large datasets with complex decision boundaries	Reduces overfitting compared to single trees, robust performance	Less interpretable, higher computational cost
Naive Bayes	Problems where feature independence assumption approximately holds	Very fast training and prediction, works well with small datasets	Strong independence assumption often unrealistic
Logistic Regression (Linear Classifier)	Linearly separable data or baseline classification tasks	Fast training, interpretable model	Limited ability to model complex nonlinear boundaries
Linear Discriminant Analysis (LDA)	Data with approximately Gaussian class distributions	Computationally efficient, provides probabilistic interpretation	Assumes equal covariance matrices across classes
AdaBoost	Object detection pipelines with simple features (e.g., Haar features)	Combines weak learners to build strong classifiers	Sensitive to noisy data and outliers

Task 3: Mean-Shift Segmentation for Colour Images

In this task, a colour image segmentation algorithm based on the mean-shift principle was implemented. The goal of the algorithm is to group pixels into homogeneous regions based on both colour similarity and spatial proximity.

For the segmentation process, each pixel was represented in a five-dimensional feature space:

$$[x, y, L, a, b]$$

where x and y denote the spatial coordinates of the pixel, and L, a, b correspond to the colour components in the CIELAB colour space.

Using both spatial and colour information is essential for meaningful segmentation. If only colour values were considered, pixels with similar colours but located far apart in the image might incorrectly be assigned to the same segment. Including spatial coordinates ensures that nearby pixels are more likely to belong to the same region.

Methodology

Each pixel was converted into a normalized feature vector:

$$\left[\frac{x}{h_s}, \frac{y}{h_s}, \frac{L}{h_r}, \frac{a}{h_r}, \frac{b}{h_r} \right]$$

where h_s represents the spatial radius and h_r represents the colour radius. These parameters control the influence of spatial proximity and colour similarity in the segmentation process.

Results

Fig. 1 shows the segmentation outputs obtained with different parameter values. Smaller spatial and colour radii preserve more image details and generate many small regions. Increasing these parameters merges neighbouring regions and produces smoother segmentation with fewer clusters.

Conclusion

From the results it can be observed that the segmentation behaviour strongly depends on the chosen parameter values. As the parameters increase, neighbouring regions merge and the segmentation becomes smoother with fewer clusters.

This behaviour confirms that the mean-shift algorithm is highly sensitive to the spatial and colour bandwidth parameters, which directly control the granularity of the segmentation.



(a) $(h_s, h_r) = (8, 12)$



(b) $(h_s, h_r) = (16, 18)$



(c) $(h_s, h_r) = (24, 28)$



(d) $(h_s, h_r) = (40, 36)$



(e) $(h_s, h_r) = (56, 45)$

Figure 1: Mean-shift segmentation results obtained using different spatial (h_s) and colour (h_r) parameters. Smaller bandwidth values produce many small segments, while larger values merge neighbouring regions and create smoother segmentation.

References

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